

$$P(A|B) = \frac{P(B|A)}{P(B)}P(A) = \frac{0.50}{0.70} \times 0.20 = 0.143$$

If the analyst uses the bankruptcy prediction model as a guide, the probability of failure declines from 20 percent to 14.3 percent.

4. In a typical year, 5 percent of all CEOs are fired for “performance” reasons. Assume that CEO performance is judged according to stock performance and that 50 percent of stocks have above-average returns or “good” performance. Empirically, 30 percent of all CEOs who were fired had “good” performance. Using Bayes’ formula, what is the probability that a CEO will be fired given “good” performance? (Hint, let $P(A)$ be the probability of a CEO being fired, $P(B)$ be the probability of a “good” performance rating, $P(B | A)$ be the likelihood of a “good” performance rating given that the CEO was fired, and $P(A | B)$ be the likelihood of the CEO being fired given a “good” performance rating.)

- A. 1.5 percent
- B. 2.5 percent
- C. 3.0 percent

Solution:

C is correct. With Bayes’ formula, the probability of the CEO being fired given a “good” rating is

$$P(A|B) = \frac{P(B|A)}{P(B)}P(A)$$

where

$P(A) = 0.05$ = probability of the CEO being fired

$P(B) = 0.50$ = probability of a “good” rating

$P(B | A) = 0.30$ = probability of a “good” rating given that the CEO is fired

With these estimates, the probability of the CEO being fired given a “good” rating is

$$P(A|B) = \frac{P(B|A)}{P(B)}P(A) = \frac{0.30}{0.50} \times 0.05 = 0.03$$

Although 5 percent of all CEOs are fired, the probability of being fired given a “good” performance rating is 3 percent.